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END SEMESTER EXAMINATION 2024-25

Name of the Course: B. Tech. PE

Semester VI

Name of the Paper: Data Science

Paper Code: TPE 605

Time: 3 Hours

Maximum Marks: 100

Note: -

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1 (20marks) **CO1**

(a) Explain the various types of Analytics i.e. Descriptive, Predictive, and Prescriptive used in Data Science. Describe their purpose, functionality, and practical applications in real-world industries such as healthcare, retail, and finance. Additionally, explain how basic descriptive analysis (including mean, median, mode, and standard deviation) is applied to datasets to summarize numerical data and derive initial insights from raw information.

(b) Describe the Analytics Methodology Framework used in Data Science and elaborate on its phases including problem definition, data collection, data preparation, model development, evaluation, and deployment. Also, explain the general process of implementing a data science solution, including steps such as data loading, identifying the problem, data cleaning, and generating summary statistics.

(c) Discuss the different types of Data Science Models and Algorithms commonly used in the field, such as regression, classification, clustering, and recommendation systems. Provide examples for each type and compare their purposes. Also, clearly differentiate between supervised and unsupervised learning approaches in terms of input data, output prediction, and real-world applications.

Q2 (20marks) **CO1**

(a) Define SELECT statements in SQL. How are SELECT statements used to retrieve specific data from a database? Explain with syntax and examples. Write an SQL query to select employee names and salaries from an "Employees" table where the salary is above 50,000.

Employees Table

EmployeeID	Name	Department	Salary
1	Rajesh Kumar	HR	48000
2	Priya Singh	IT	52000
3	Anil Mehra	Finance	55000
4	Divya Patel	Marketing	47000
5	Ravi Verma	IT	60000

(b) What are different types of SQL JOINs? Explain INNER JOIN, LEFT JOIN, RIGHT JOIN, and FULL OUTER JOIN with examples. Write SQL queries using JOINs to extract data from two tables: `Orders` and `Customers`, such that you get a list of customer names and their corresponding order amounts.

Customers Table

CustomerID	CustomerName
101	Aman Gupta
102	Neha Sharma
103	Rohan Das
104	Sneha Roy

Orders Table

OrderID	CustomerID	OrderAmount
1	101	2000
2	103	3500
3	101	1500
4	105	4000

(c) Explain the purpose and working of GROUP BY and HAVING clauses in SQL with examples. How do they differ from WHERE? Write an SQL query to group products by category in a "Products" table and find the average price of products in each category, filtering out groups where the average price is less than 100.

Products Table

ProductID	ProductName	Category	Price
1	Pen	Stationery	20
2	Notebook	Stationery	80
3	Calculator	Electronics	500
4	Laptop	Electronics	45000
5	Desk Lamp	Furniture	700
6	Chair	Furniture	1500

Q3 (20marks) **CO2**

(a) Define Descriptive Statistics. Explain mean, median, mode, variance, and standard deviation with examples. Calculate all five measures manually for the following dataset:

Sample Dataset:
45, 67, 89, 45, 70, 90, 55, 67, 80, 55

(b) (i) Describe the Analytics Methodology Framework and discuss how it helps in structured problem-solving for data-driven projects.

(ii) Task Summary for Python Program:

1. Load the dataset using pandas
2. Identify missing or incomplete data
3. Clean the data (e.g., fill missing scores with average, drop unnamed students)
4. Output a summary report:
   - Number of students
   - Average Math and English scores
   - Gender distribution
   - Students with less than 75% attendance

Sample Dataset: students.csv

StudentID	Name	Age	Gender	MathScore	EnglishScore	Attendance (%)
1	Aditi Sharma	18	F	88	92	95
2	Raj Verma	17	M	76	81	90
3	Simran Kaur		F	67		85
4	Aman Singh	19	M		74	70
5	N/A	18		91	89	100

(c) Using Microsoft Excel, perform the following data visualization tasks on the given dataset:

1. Create a Bar Chart to show the average scores of students in each subject.
2. Create a Histogram for Math scores to understand the distribution of marks.
3. Create a Box Plot (Box and Whisker Chart) for English scores to visualize the spread and detect any outliers.

Dataset: Use the table below to perform all visualizations.

StudentID	Name	MathScore	EnglishScore	ScienceScore
1	Aditi Sharma	88	92	85
2	Raj Verma	76	81	78
3	Simran Kaur	67	74	70
4	Aman Singh	90	79	88
5	Pooja Rawat	82	85	83
6	Ravi Mehra	60	70	65
7	Neha Joshi	95	93	90

Q4 (20 marks) **CO3**

(a) Explain the data processing capabilities of R. What are vectors, data frames, and lists in R and how are they used? Write an R script to create a data frame, manipulate columns (filter, add new ones), and print summary statistics.

(b) Write an R program to implement simple linear regression using the following dataset. Use the model to predict marks based on the number of study hours. Also, interpret the regression output including the coefficients, R-squared value, and significance (p-value) of the model.

Sample Dataset: study_hours.csv

StudentID	StudyHours	Marks
1	1.5	42
2	2.0	50
3	2.5	55
4	3.0	61
5	3.5	67
6	4.0	72
7	4.5	78
8	5.0	85
9	5.5	88
10	6.0	95

(c) What is Time Series Forecasting? Explain components of time series (trend, seasonality, etc.) and mention common forecasting models.

Q5 (20 marks) **CO4**

(a) What is Machine Learning? Explain its types with examples (Supervised, Unsupervised, and Reinforcement Learning). Using scikit-learn, build and evaluate a classification model (e.g., Decision Tree or KNN) on a labeled dataset like Iris or Titanic.

(b) Explain why preparing and cleaning data is a critical step before applying algorithms. Describe the role of handling missing values, removing duplicates, converting data types, and normalizing numerical values in ensuring model accuracy and reliability. Also, elaborate on commonly used data manipulation functions in the pandas library and their theoretical purpose for dropna(), fillna(), duplicated(), astype(), apply(), normalize().

(c) (i) Describe different visualization libraries in Python and how visualizations assist in EDA (Exploratory Data Analysis).
(ii) Use Python's `Seaborn` and `Matplotlib` libraries to generate a correlation heatmap and pair plot for a multivariate dataset. Sample Dataset: stud_performance.csv

StudentID	MathScore	ReadingScore	WritingScore	StudyHours	Attendance
1	88	92	90	5	95
2	76	81	80	4	88
3	67	74	73	3.5	85
4	90	89	91	5.5	97
5	82	85	83	4.5	90
6	60	70	68	2.5	80
7	95	94	96	6	98
8	78	79	77	4	87